National Aeronautics and Space Administration

EEBIOMASS Workshop 2026

Scalable EO Data Access on the MAAP

Finding Where NISAR and ESA BIOMASS Meet

Project: PP0104807 - Opportunities in Data Science Exploration of ESA Biomass Data with NASA MAAP

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NASA MAAP MSFC Data Team, ESA, NASA HQ, JPL and Collaborators

Takeaways :

- An end-to-end cloud-native platform for open EO science
- Where NISAR and BIOMASS co-cover your study area
- A list of matching granule and item IDs, ready for data access
- A repeatable pattern for any pair of products

For more information, reach out on:

- NASA MAAP - support@maap-project.org
- ESA MAAP - https://portal.maap.eo.esa.int/biomass/initiative-membership/

Outline

Part 0 About MAAP

Part 1 The (Data) Science Question

Part 2 Live walkthrough

PART 0

Multi - mission Algorithm and Analysis Platform

MAAP: A cloud- native platform for Earth Observation Open Science

Who MAAP Serves

MAAP currently serves multiple user communities

Science PIs & Teams

- **Researchers** working with NASA Earth science data who need scalable compute
- **Collaborative** groups (GEOTREES, ABoVE) needing shared workspaces
- No in-house expertise to configure or manage compute clusters
- **Constrained budgets** — cannot stand up dedicated infrastructure

Mission Data Producers

- Active usage by **SMAP, TROPESS** teams
- Mission data processing heritage used across **NISAR, SWOT, OPERA, SMAP, OCO**
- Needs data and compute **isolation** from other teams
- Wants pre-defined ingest pipelines into established archives
- No platform lock-in: data and algorithms stay portable

International Partners

- ESA and NASA co-developed platform from inception
- International teams (e.g. GEOTREES) **collaborate** directly on **shared** infrastructure
- Data **fusion** support between **NISAR, BIOMASS, and GEDI** data sets
- Joint **authentication**: NASA Earthdata + ESA EOIAM

Key Point: One single platform serving researchers, missions, and international partners.

MAAP Use Cases Today

Science-driven platform supporting individual researchers and mission teams

Biomass & Carbon

NISAR, GEDI, BIOMASS, ICESat-2 & Landsat data analysis for global biomass estimation

SMAP Algorithm Team Support

Active usage by SMAP algorithm team for fusion data product prototyping.

Fire Monitoring (FEDS)

Fire Event Data Suite ingests satellite data to track active fires in near-real-time

ABoVE Campaign

Arctic-Boreal studies performing aboveground biomass mapping

CARDAMOM

Carbon Data Assimilation Model Framework assimilating multiple satellite observations with field measurements

ESA Collaboration

Joint NASA-ESA platform since inception — international teams (e.g. GEOTREES) collaborate directly

Key Point: Already powering real science across biomass, fire, carbon, and climate.

Platform Features



Analysis Environment

JupyterHub on Science Cloud (2i2c-managed) with custom MAAP extensions, ML integration, Dask integration, and collaborative notebooks



Data Processing System (DPS)

Scalable algorithm execution via OGC Application Packages — register, run, and chain workflows at production scale. Built with same HySDS processing system used on flagship missions such as NISAR, SWOT, OPERA, SMAP, and OCO-2 & 3.



Data Layer

STAC-based multi-tenant and multi-catalog including NASA CMR and ESA integration — co-located multi-mission data (GEDI, ICESat-2, BIOMASS, NISAR, and more)
Browsable and customizable, web and api interfaces.



Dynamic Visualization

Raster, multi-dimensional, and Vector based dynamic tiling and mosaic solutions on standardized APIs. Add any cataloged data as layer to web, notebook and desktop maps. Scrolly-telling widgets for adding maps to websites.



API & Interoperability

MAAP API + maap-py Python client — joint NASA-ESA auth, cross-platform algorithm sharing, OGC standards throughout.

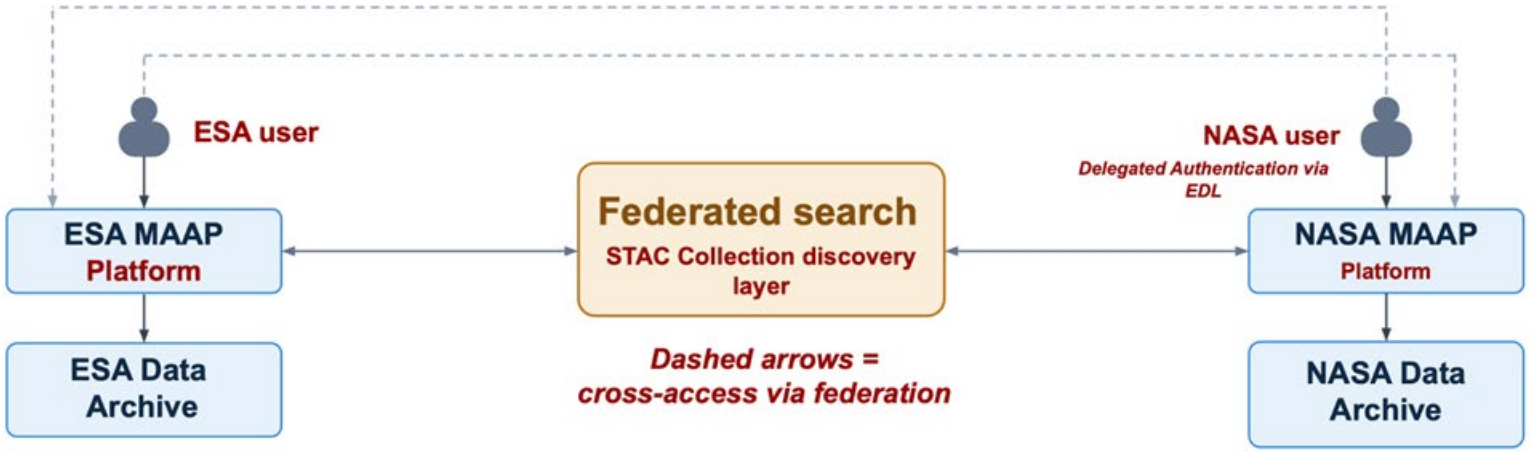
Why This Matters

- Unified computing environment - *already built and running*
- Multisource data harmonization – e.g. *across GEDI, ICESat-2, BIOMASS & NISAR*
- Cross-team collaboration - *proven with NASA-ESA joint*
- Algorithm dev + at-scale processing - *from prototype to production in one platform*
- Open science compliance - *all data, algorithms, software are open access*

Key Point: Collaboration and interoperability through scalable compute, multi-source data, API-driven visualization capabilities.

What MAAP offers? Data Access

- One platform co - developed by NASA and ESA
- One search spans NASA CMR, NASA MAAP STAC and ESA MAAP STAC (federated search)
- One login: NASA Earthdata Login also reaches ESA data (delegated authentication)
- Runs in the cloud next to the data: no local downloads or AWS setup
- Built for fusion: NISAR, BIOMASS and GEDI side by side



Federated Search

Data Interoperability

Use free-text search and spatial/temporal filters to find collections across multiple STAC APIs and get code hints for item-level access



Figure: Visualization of the 30m vegetation height ICESAT v3.1 product on the STAC Collection Discovery Interface

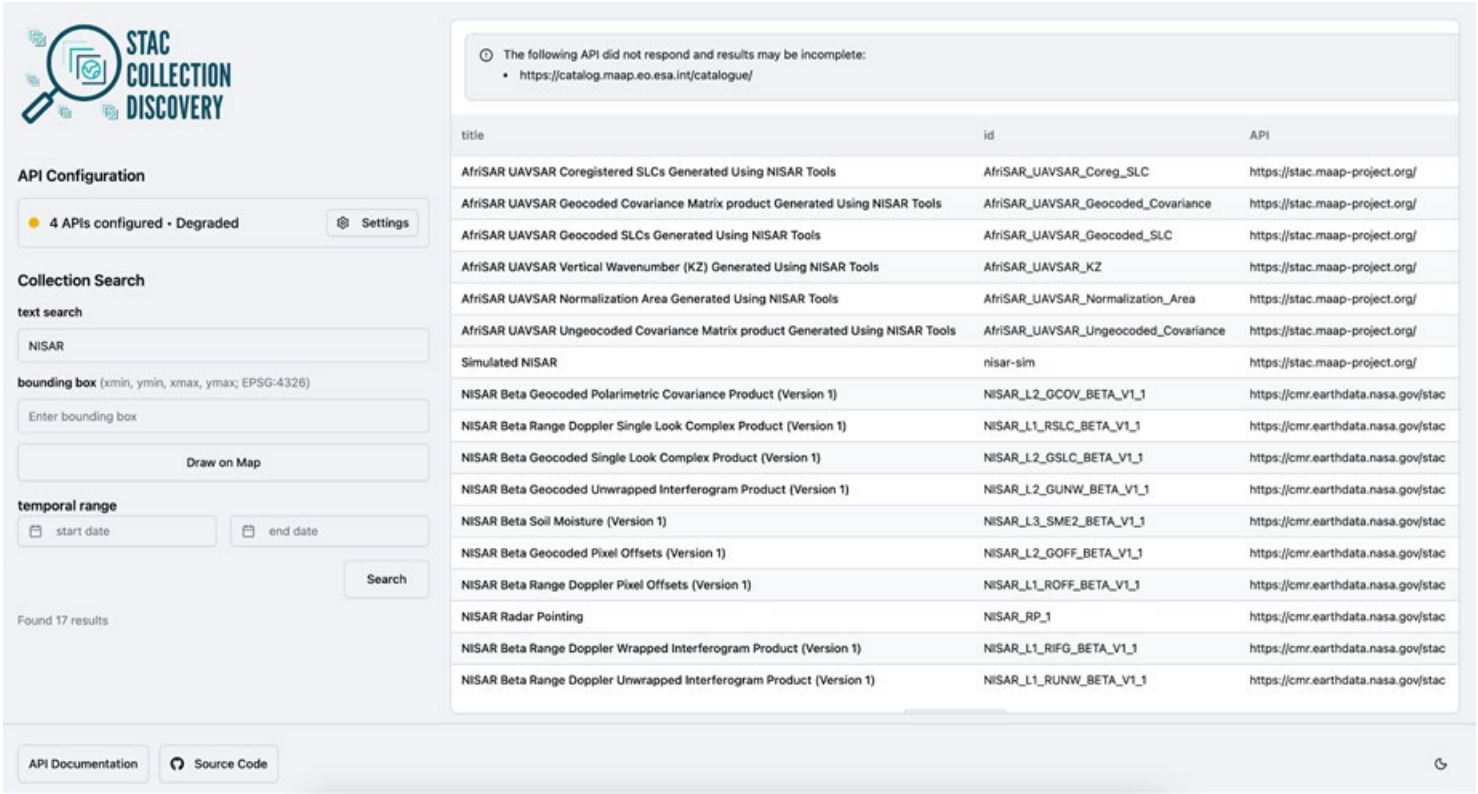


Figure: List of all collections available from different sources based on the search query "NISAR"

Scaling and Open Science Support

Data Interoperability

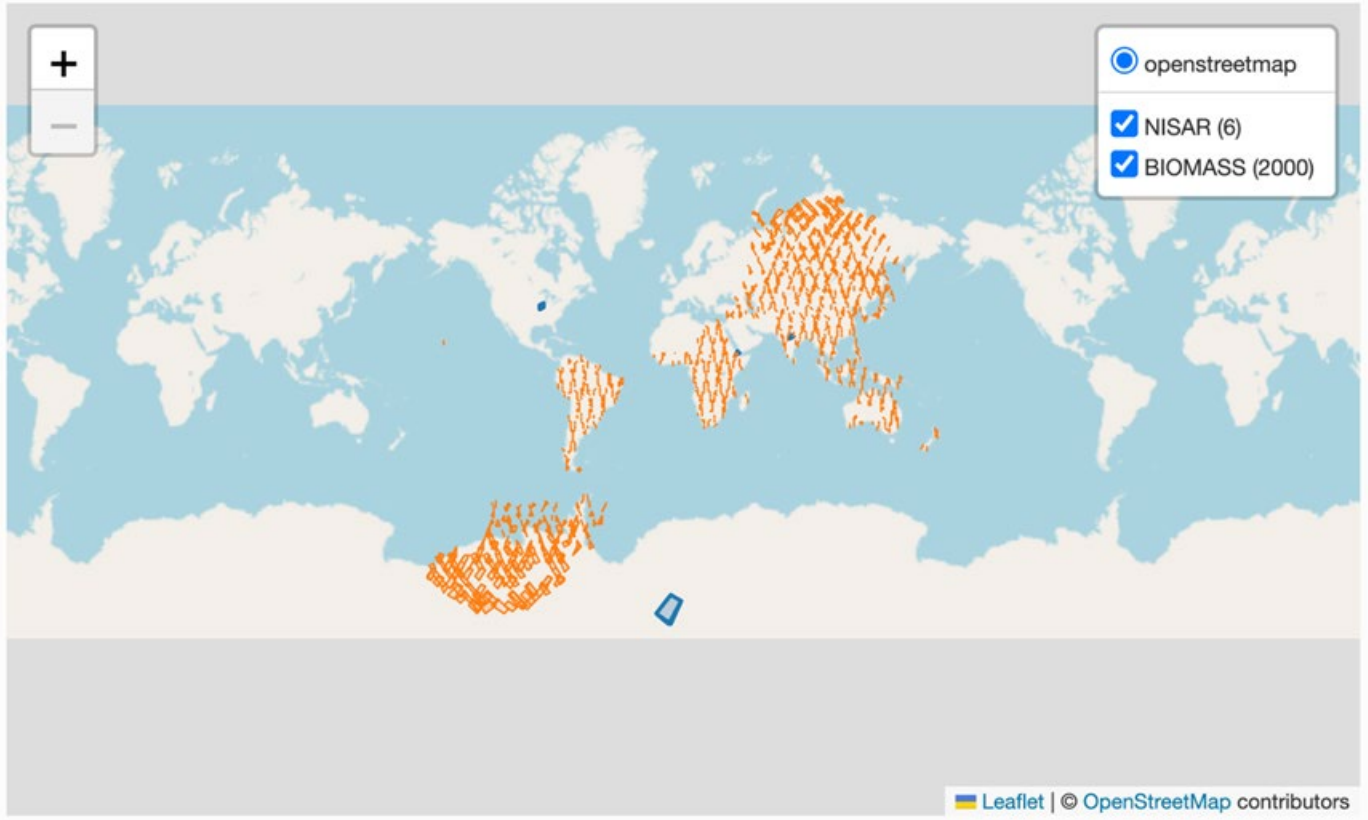
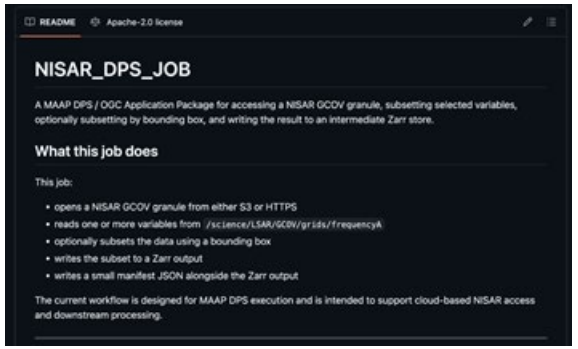
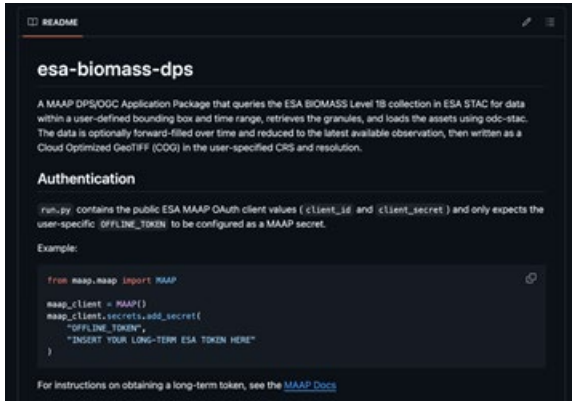


Figure: Visualizations showing overlap of NISAR and BIOMASS products for data-fusion use case and OGC App packages for scalable data access for DPS job submissions

- OGC Application Packages for scalable data access for DPS processes, useful for data-fusion use cases
- Standardizing DPS outputs in structure and naming conventions





Interactive Web Map

Data Interoperability



Figure: Biomass Dashboard

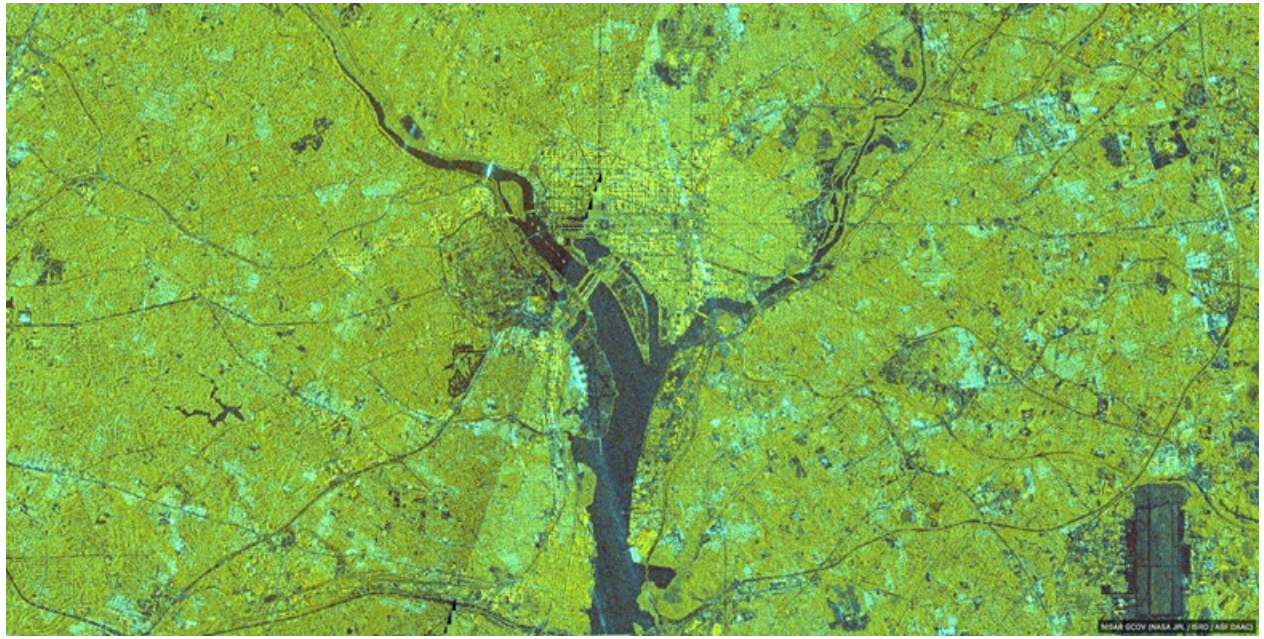


Figure: Cloud-optimized reads of NISAR layers tiled interactively by TiTiler-CMR.

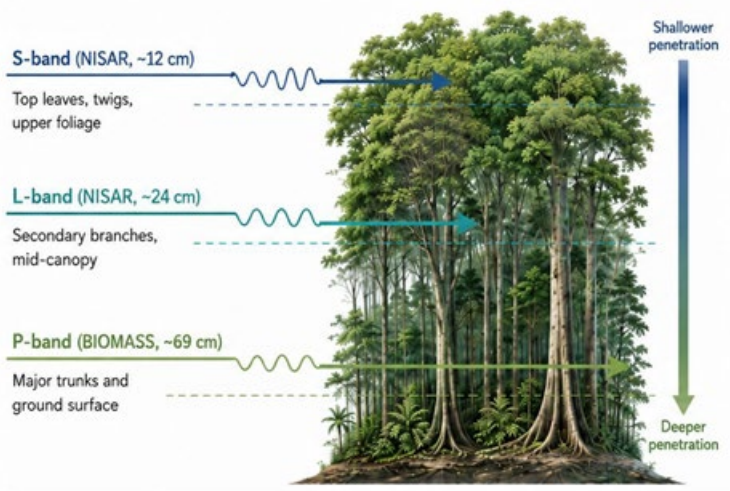
PART 1

The (Data) Science Question

“Where do NISAR and BIOMASS cover the same ground?”

Meet Our Research

- Studies forests and above-ground biomass on MAAP
- Wants to combine NISAR(NASA–ISROSAR)with ESABIOMASSdata
- Problem: the two datasets live in different archives, catalogs and logins
- First question : where and when do they actually overlap?



How deep each radar band sees into a forest: short S-band waves stop in the upper foliage, NISAR's L-band reaches branches and mid-canopy, BIOMASS's P-band reaches the trunks and the ground.

NISAR: One Satellite, Two Radars

- L-band radar (~24 cm wavelength), built by NASA/JPL
  - Sees through clouds, day or night, and into forest canopy
- S-band radar (~10 cm wavelength), built by ISRO
  - More sensitive to crops and lighter vegetation
- Covers the globe on a 12-day repeat
- This demo uses L-band data



Artist's concept of NISAR in orbit. The large golden drum is the 12 m radar reflector, which unfolds in space like an umbrella; the solar panels power the two radars.

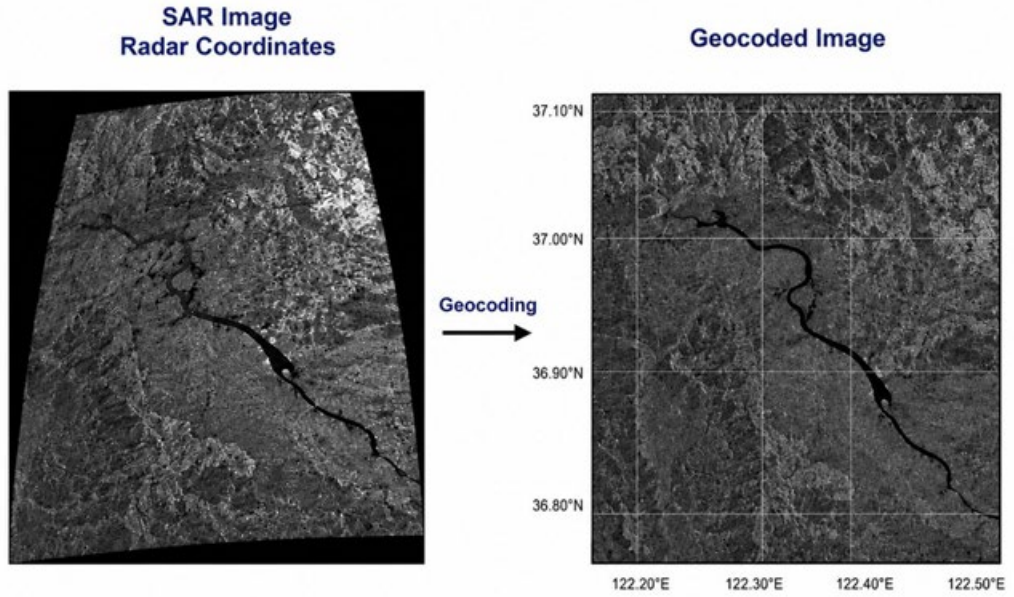
Image: NASA/JPL-Caltech (public domain), via Wikimedia Commons

NISAR Products: What We Used and Other Choices

We used: Level-2 GCOV (Geocoded Polarimetric Covariance)
• Backscatter on a map grid, the easiest product to compare with other datasets

Other choices:
• Level 1 (radar geometry): RSLC, RIFG, RUNW, ROFF
• Level 2 (map geometry): GSLC, GCOV, GUNW, GOFF
• Level 3: SME2 (soil moisture)

Quality tiers:
• BETA: pre-calibration
• PROVISIONAL: calibrated (used here, from June 17, 2026 onward)



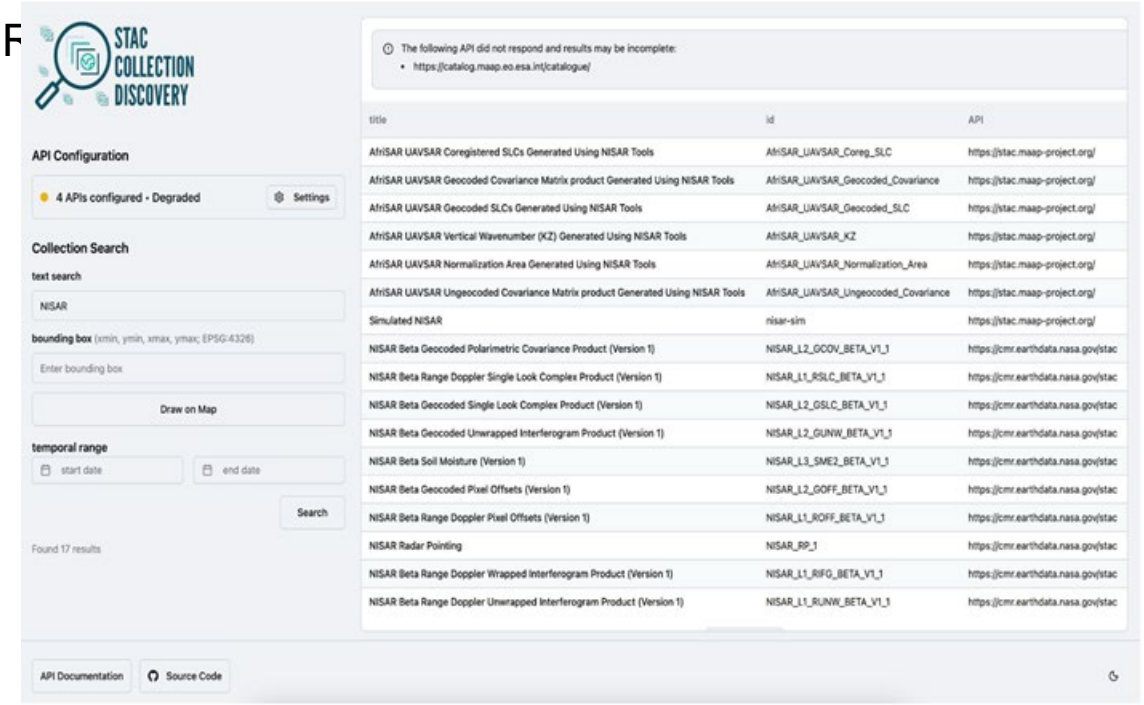
The same scene before and after geocoding. Left: Level-1 image in radar coordinates (distorted, not north-up). Right: Level-2 image on a map grid with latitude and longitude, ready to overlay on a map.

Where NISAR Lives and How to Access It

- L-band products are archived at NASA's ASF DAAC (Alaska Satellite Facility)
- Stored in the AWS cloud (us-west-2), cataloged in NASA CMR
- Log in with NASA Earthdata Login (EDL)
- Two ways in:
  - Inside MAAP: direct S3 streaming (fastest)
  - Outside MAAP: HTTPSstreaming or download
- Search and open with earthaccess

MAAP Docs: NISAR Access and Exploration

ASF NISAR Data User Guide



List of all collections available from different sources based on the search query "NISAR"

ESA BIOMASS: The First P- band Radar in Space

- P- band radar (~70 cm wavelength): the longest wavelength in orbit
- Reaches trunks and branches in dense forests, which is where biomass lives

We used: Level - 1b DGM (Detected Ground Multi - look)
- Analysis - ready backscatter images

Other choices:
- Level- 1a: Single look Complex Slant (SCS)
- Level- 1c: Co- registered single - look complex stacks
- Level- 2a: Forest retrieval products



Artist's view of the BIOMASS satellite. Its P-band radar penetrates the canopy and measures trunks and branches, where most of the forest's wood, and therefore carbon, is stored.

Image: NASA/JPL-Caltech (public domain), via Wikimedia Commons

Where BIOMASS Lives and How to Access It

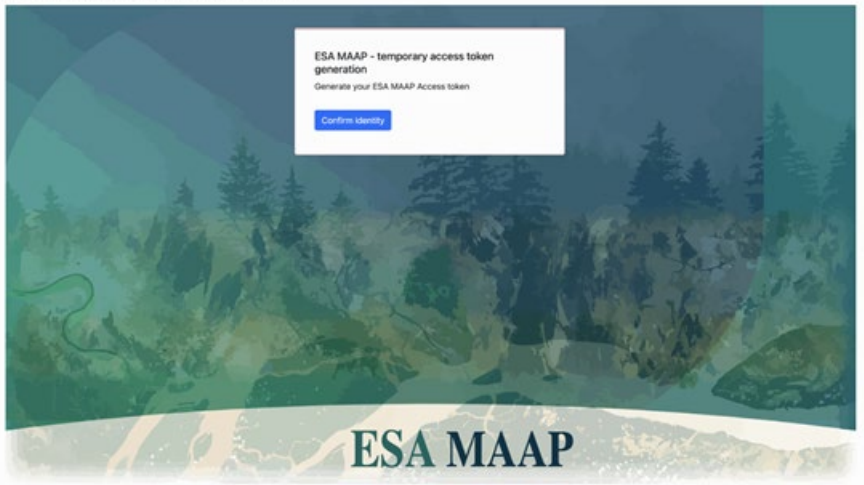
- Hosted by ESA on the ESA MAAP catalog (STAC)
  https://catalog.maap.eo.esa.int/catalogue/

- Searching the catalog is open, no token needed

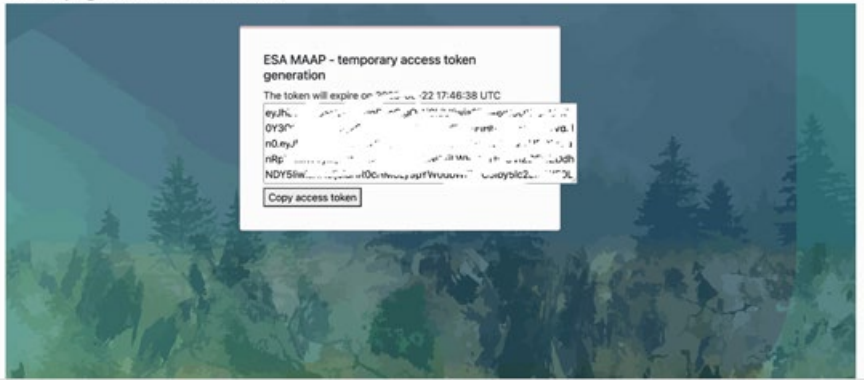
- Reading or downloading the data needs a token:
  1. Log in to the ESA MAAP portal with your NASA Earthdata Login
  2. Copy your offline token (valid 90 days)
  3. Exchange it for a short - lived access token in your notebook

MAAP Docs: ESA BIOMASS Data Access and Visualization

- NASA EDL authorization screen:



- Token page after authorization:



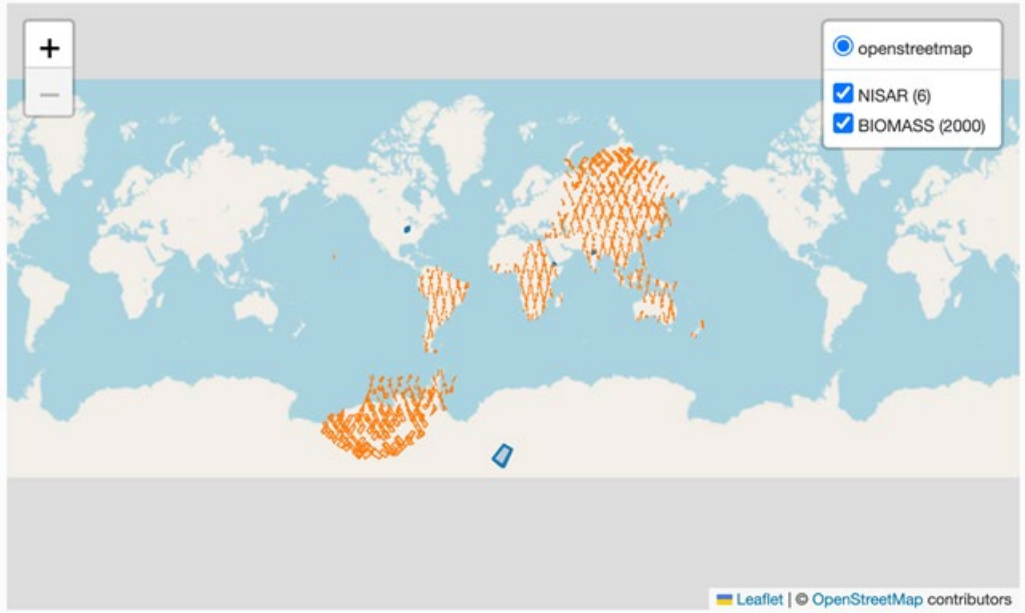
ESA MAAP temporary access token workflow showing NASA Earthdata Login (EDL) authorization and token generation after successful authentication.

What "Overlap" Means Here

• Every NISAR granule and BIOMASS item has a footprint: its outline on the map

• Overlap = a NISAR outline and a BIOMASS outline intersect

• Result: a list of matching NISAR ☻ BIOMASS pairs and a map of the shared areas



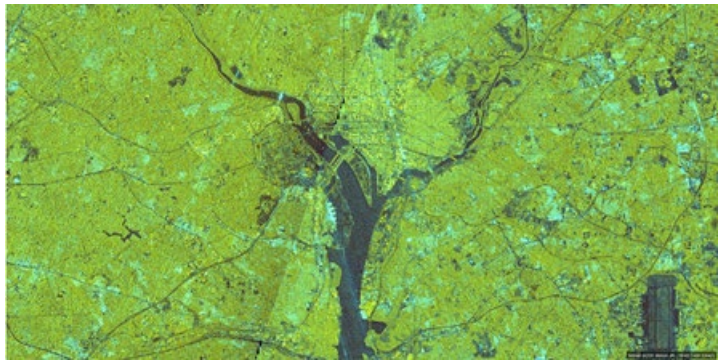
Footprints of NISAR GCOV granules (blue, first 500) and BIOMASS Level-1b products (orange, first 2,000) on an interactive map.
Fusion is only possible where blue and orange overlap

Look Before You Download: Cloud - Native and Cloud - Optimized

Cloud-native: bring the analysis to the data
• NISAR data sits in AWS; MAAP runs in the same cloud region
• No copying terabytes to your laptop

Cloud-optimized: read only the part you need
• Files are laid out for partial reads over the network (chunked HDF5, GeoTIFF)
• Example: a Nepal flood map on MAAP read ~108 MB of NISAR data instead of 14 GB

This demo goes one step lighter:
• We search metadata only (IDs, dates, footprints)
• Nothing is downloaded until you know which scenes overlap



Cloud-optimized reads of NISAR layers tiled interactively by TiTiler-CMR.
Development Seed, *TiTiler-CMR – Visualizing orbital swath data: NISAR GCOV in India*

The Workflow at a Glance



Define → Search → Map → Intersect → Focus

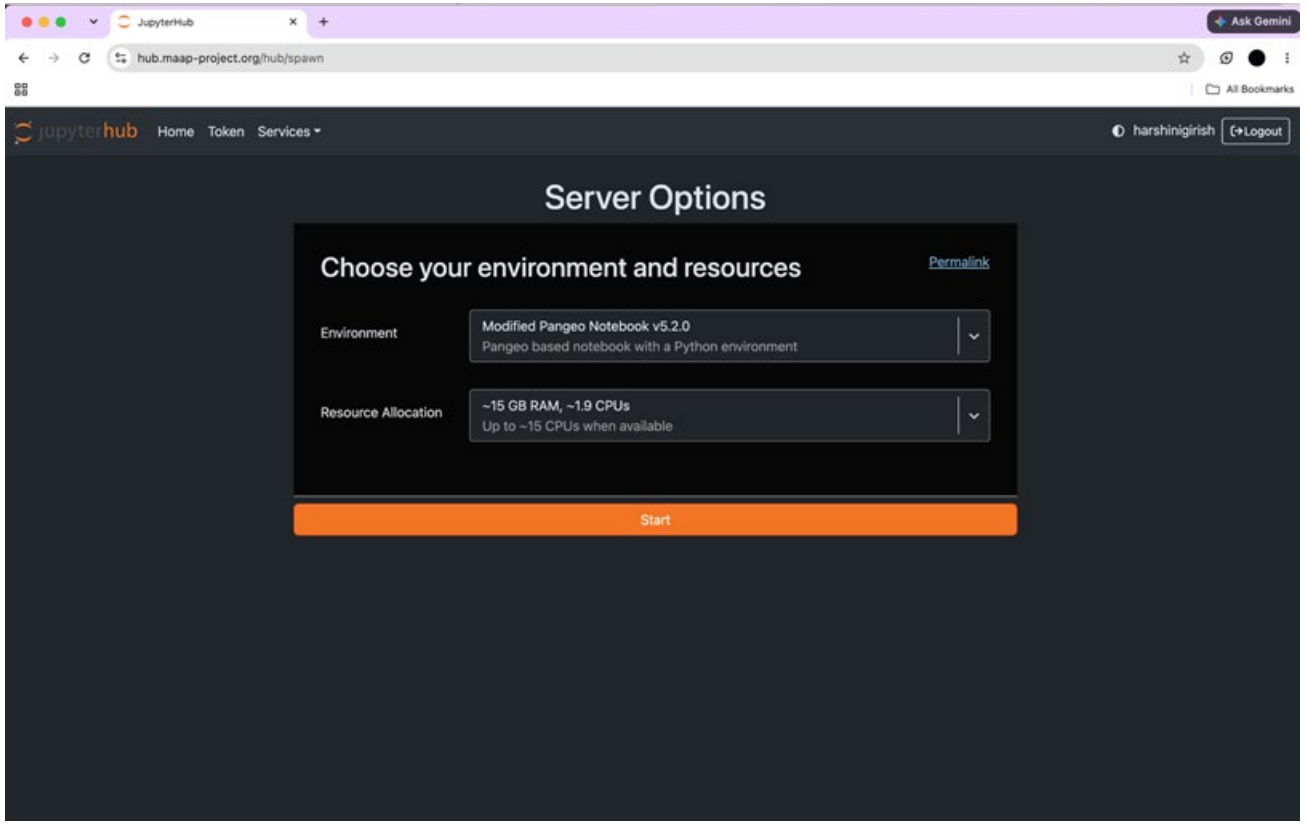
PART 2

Live Walkthrough

Six steps — now in the MAAP notebook

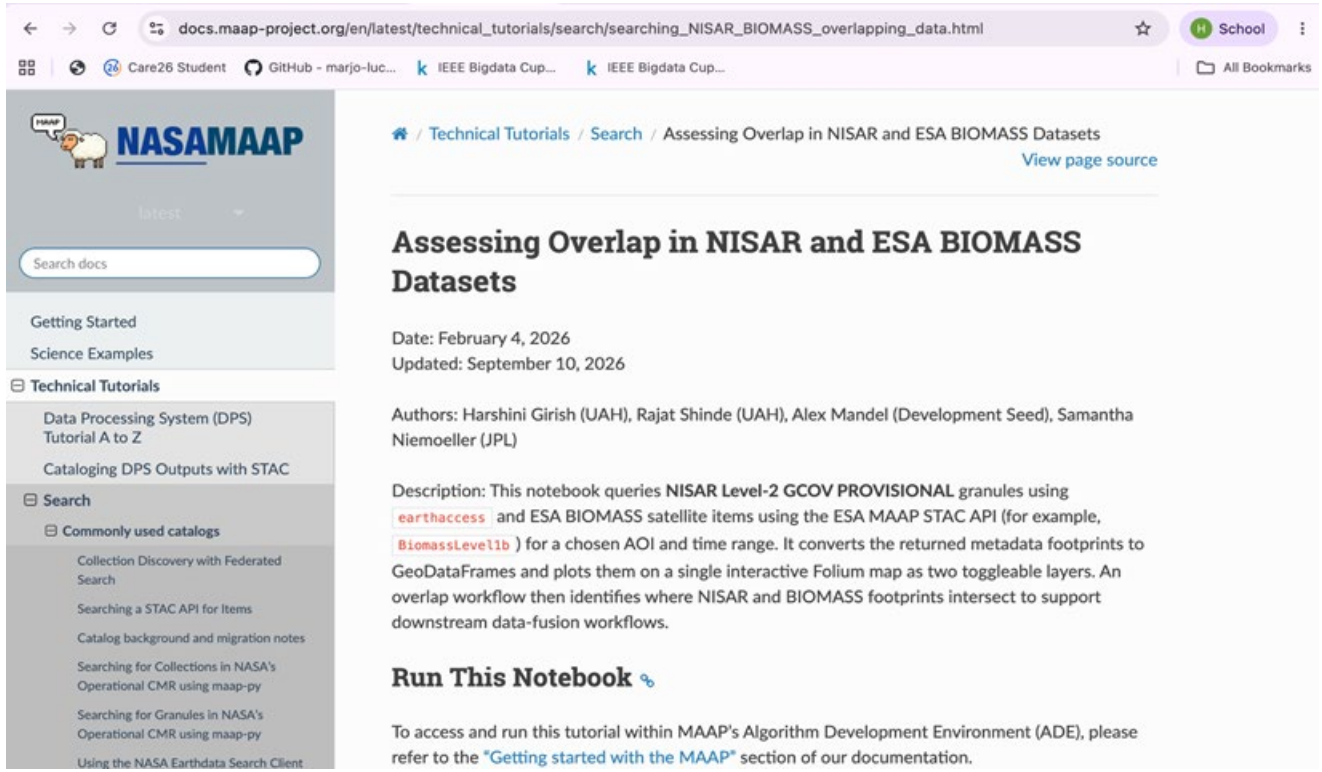
Open the Tutorial in MAAP

- Log in to the MAAP HUB and start a Pangeo v5 workspace



Launching a Pangeo workspace in the MAAP hub

- Open MAAP Docs: Search › Assessing Overlap in NISARand ESABIOMASS



Opening the MAAP technical tutorial, "Assessing Overlap in NISAR and ESA BIOMASS Datasets," from the MAAP documentation.

Set Your Search Window

- BBOX: Study area (keep it small)
- NISAR_DT/ BIOMASS_DT: Date ranges for each mission
- NISAR_SHORT_NAME: NISAR_L2_GCOV_PROVISIONAL_V1
- BIOMASS_STAC_URL: ESA MAAP catalog address
- BIOMASS_COLLECTION: BiomassLevel1b
- NISAR_MAX_ITEMS/ BIOMASS_MAX_ITEMS: Result limits for faster searches

[3]:
```
# Common query parameters (edit to your AOI/time window)
# A smaller AOI is strongly recommended. The global BBOX is retained only as a generic example.
BBOX = [-180, -90, 180, 90]                    # [min_lon, min_lat, max_lon, max_lat]

# PROVISIONAL products are available from 2026-06-17 onward, plus select earlier
# acquisitions that have been reprocessed to PROVISIONAL quality.
NISAR_DT = ("2026-06-17", "2026-09-10")
NISAR_SHORT_NAME = "NISAR_L2_GCOV_PROVISIONAL_V1"
NISAR_MAX_ITEMS = 500                          # return at most the first 500 CMR results

BIOMASS_DT = "2024-01-01/.."                    # adjust if needed
BIOMASS_STAC_URL = "https://catalog.maap.eo.esa.int/catalogue/"
BIOMASS_COLLECTION = "BiomassLevel1b"
BIOMASS_MAX_ITEMS = 200                        # keep broad STAC searches manageable
```

Search inputs: area, dates, collections, limits

Query NISAR with earthaccess

- earthaccess .search_data() asks NASA CMR for matching granules
- Each granule's footprint becomes a row in a GeoDataFrame
- Check the count and preview the first rows
- Sample run: 500 granules

[3]:

	nisar_id	geometry
0	NISAR_L2_PR_GCOV_023_040_D_124_4005_SHSH_A_202...	MULTIPOLYGON (((-103.79052 -71.31863, -104.850...
1	NISAR_L2_PR_GCOV_023_040_D_125_4005_SHSH_A_202...	MULTIPOLYGON (((-104.73943 -72.34834, -107.090...
2	NISAR_L2_PR_GCOV_023_040_D_126_4005_SHSH_A_202...	MULTIPOLYGON (((-106.95305 -74.39485, -109.855...
3	NISAR_L2_PR_GCOV_023_040_D_127_4005_SHSH_A_202...	MULTIPOLYGON (((-109.68257 -76.42305, -113.520...
4	NISAR_L2_PR_GCOV_023_040_D_128_4005_SHSH_A_202...	MULTIPOLYGON (((-113.28891 -78.4149, -118.3023...

NISAR granule footprints as a GeoDataFrame

Query BIOMASS with STAC

- pystac-client opens the ESA MAAP STAC endpoint
- Search BiomassLevel1b with the same box and time window
- Footprints go into a second GeoDataFrame
- Sample run: 200 items (hit the cap)

https://portal.maap.eo.esa.int/ini/services/auth/token/

BIOMASS items returned: 200

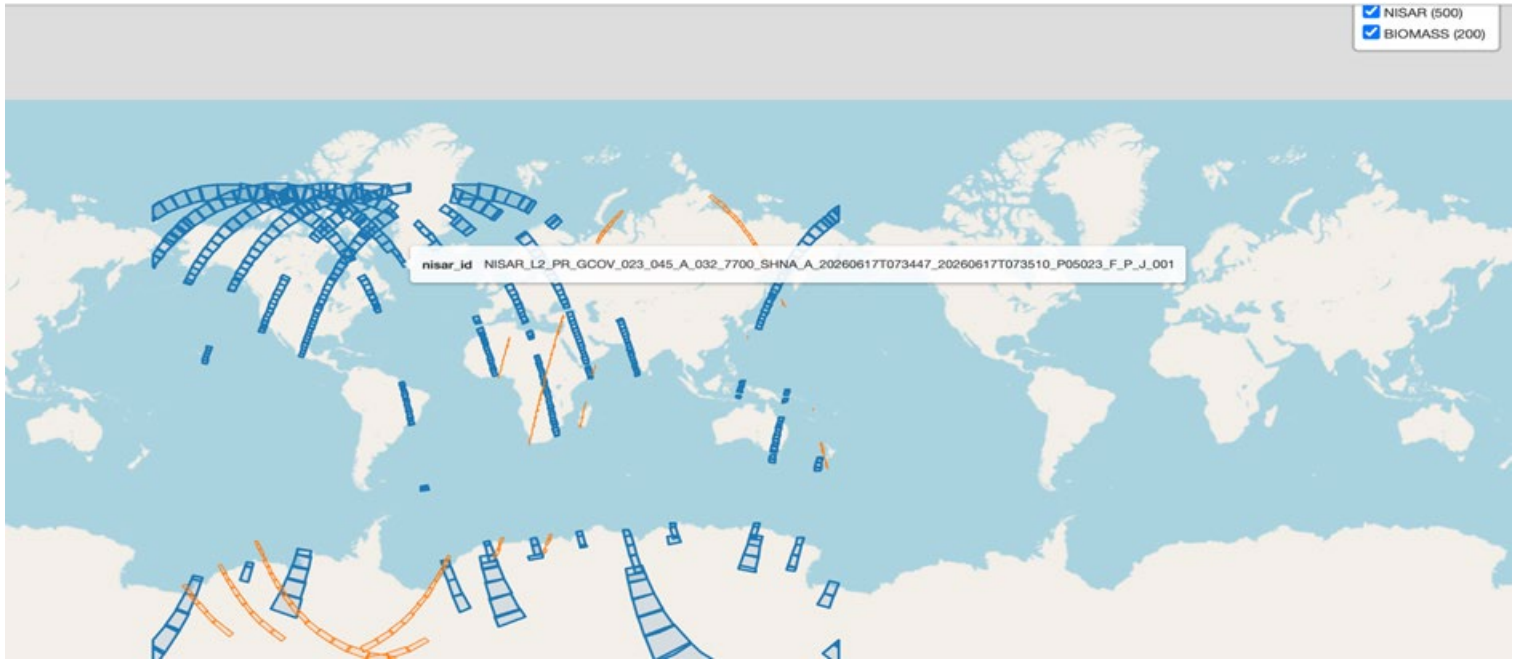
[4]:

	biomass_id	start_datetime	end_datetime	datetime	geometry
0	BIO_S1_DGM__1S_20260927T204317_20260927T204335...	2026-09-27T20:43:17.908Z	2026-09-27T20:43:35.373Z	2026-09-27T20:43:17.908Z	POLYGON ((131.84128 29.58142, 131.22783 29.446...
1	BIO_S1_DGM__1S_20260927T191839_20260927T191905...	2026-09-27T19:18:39.802Z	2026-09-27T19:19:05.578Z	2026-09-27T19:18:39.802Z	POLYGON ((113.05752 76.09779, 111.56222 75.677...
2	BIO_S1_DGM__1S_20260927T191820_20260927T191841...	2026-09-27T19:18:20.840Z	2026-09-27T19:18:41.090Z	2026-09-27T19:18:20.840Z	POLYGON ((117.4407 75.05116, 115.94648 74.6612...
3	BIO_S1_DGM__1S_20260927T191801_20260927T191822...	2026-09-27T19:18:01.879Z	2026-09-27T19:18:22.128Z	2026-09-27T19:18:01.879Z	POLYGON ((120.44418 74.19271, 118.95983 73.822...
4	BIO_S1_DGM__1S_20260927T191742_20260927T191803...	2026-09-27T19:17:42.916Z	2026-09-27T19:18:03.165Z	2026-09-27T19:17:42.916Z	POLYGON ((123.14616 73.29844, 121.67734 72.944...

BIOMASS footprints as a GeoDataFrame

Map the Footprints

- Folium map with a NISAR layer and a BIOMASS layer
- Layer control to toggle each mission
- Tooltips show granule and item IDs



NISAR (blue) and BIOMASS (orange) footprints

Compute and Map the Overlap

- gpd.sjoin(..., predicate="intersects") finds matching pairs
- Intersection geometry gives the shared polygons

```
Overlap pairs: 29
[10]:
```

	nisar_id	biomass_id
0	NISAR_L2_PR_GCOV_023_040_D_127_4005_SHSH_A_202...	BIO_S1_DGM__1S_20260927T151835_20260927T151855...
1	NISAR_L2_PR_GCOV_023_040_D_127_4005_SHSH_A_202...	BIO_S1_DGM__1S_20260927T151854_20260927T151915...
2	NISAR_L2_PR_GCOV_023_040_D_128_4005_SHSH_A_202...	BIO_S1_DGM__1S_20260927T151757_20260927T151817...
3	NISAR_L2_PR_GCOV_023_040_D_128_4005_SHSH_A_202...	BIO_S1_DGM__1S_20260927T151816_20260927T151836...
4	NISAR_L2_PR_GCOV_023_040_D_128_4005_SHSH_A_202...	BIO_S1_DGM__1S_20260927T151835_20260927T151855...
5	NISAR_L2_PR_GCOV_023_040_D_128_4005_SHSH_A_202...	BIO_S1_DGM__1S_20260927T151854_20260927T151915...
6	NISAR_L2_PR_GCOV_023_042_D_127_7700_SHNA_A_202...	BIO_S1_DGM__1S_20260927T183457_20260927T183517...
7	NISAR_L2_PR_GCOV_023_042_D_127_7700_SHNA_A_202...	BIO_S1_DGM__1S_20260927T183516_20260927T183543...
8	NISAR_L2_PR_GCOV_023_042_D_128_7700_SHNA_A_202...	BIO_S1_DGM__1S_20260927T183457_20260927T183517...

Sample run: 29 overlap pairs

- Second map shows only the overlap areas
- Sample run: 29 overlap pairs



Overlap zone: NISAR ∩ BIOMASS over Central Africa

Takeaways :

- An end-to-end cloud-native platform for open EO science
- Where NISAR and BIOMASS co-cover your study area
- A list of matching granule and item IDs, ready for data access
- A repeatable pattern for any pair of products

For more information, reach out on:

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National Aeronautics and Space Administration

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Thank you

